

Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks (Reimers and Gurevych, 2019)

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Reimers and Gurevych (2019) present Sentence-BERT (SBERT), a modification of the BERT network using siamese and triplet networks that they claim is able to derive semantically meaningful sentence embeddings. Once you have read the paper, consider the following questions.

1. For a pair regression task, the standard BERT set-up requires pairs of sentences to be presented as input to the encoder network. Why is this set-up unfeasible if you want to find the most similar sentences in a collection?
2. What have other researchers done to overcome this problem with using BERT?
3. What is the SBERT strategy?
4. What do you understand by the term Siamese network structure?
5. What are the different pooling strategies that the authors experiment with? What works best?
6. Outline the 4 evaluation tasks used for semantic textual similarity.
7. Why would Spearman's rank correlation coefficient be better than Pearson's product-moment correlation coefficient when comparing ratings of semantic textual similarity?
8. What are the different objective functions that the authors experiment with? When is each used?
9. How is the SentEval evaluation different to the previous evaluation experiments?
10. What are the main conclusions of the paper? Are you convinced?

References

Nils Reimers and Iryna Gurevych. 2019. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*.